

Hotel Booking Data Analytics Project Report

1. Project Title

Hotel Booking Data Analytics

2. Project Overview

This project performs a comprehensive data analysis on a hotel booking dataset comprising 87,230 reservation records across City Hotel and Resort Hotel properties. The analysis investigates critical operational and revenue drivers within the hospitality business context, including cancellation behavior across lead time windows, room reassignment dynamics and associated Average Daily Rate (ADR) impact, guest engagement through special requests and parking requirements, loyalty patterns comparing repeat versus first-time guests, and geographical cancellation variation across key origin markets. The purpose of the project is to convert raw booking data into actionable business intelligence to support revenue management, operational planning, and targeted guest communication strategies.

3. Objective

The objective of this project is to use descriptive data analysis and visualization techniques to uncover meaningful patterns in hotel booking behavior. Specifically, the analysis aims to:

- Evaluate how reservation lead times correlate with cancellation likelihood.
- Measure the frequency of room reassignments and analyze whether reassigned stays yield lower or higher ADRs compared to non-reassigned stays.
- Determine if guest engagement (measured by special requests and parking space requirements) relates to booking completion rates.
- Compare repeat guests against first-time guests regarding lead times, historical booking patterns, and cancellation rates.
- Identify high-risk origin markets by evaluating cancellation rates among international guest source countries.

4. Dataset

The analysis is based on the Hotel Booking Demand dataset originally sourced from Kaggle.

Kaggle Original Source: [Hotel Booking Demand Dataset \(Kaggle\)](#)

Dataset Structure

- **Total Rows:** 87,230
- **Total Columns:** 37

Data Dictionary Summary

Column Name	Data Type	Description
hotel	Categorical (String)	Property type: City Hotel or Resort Hotel.
is_canceled	Binary (Int)	Booking status: 1 = Canceled, 0 = Not

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Column Name	Data Type	Description
		Canceled.
lead_time	Numerical (Int)	Number of days between booking date and arrival date.
arrival_date_year	Numerical (Int)	Year of arrival date (2015-2017).
arrival_date_month	Categorical (String)	Month of arrival date.
arrival_date_week_number	Numerical (Int)	Week number of arrival date.
arrival_date_day_of_month	Numerical (Int)	Day of the month of arrival date.
stays_in_weekend_nights	Numerical (Int)	Number of weekend nights (Saturday or Sunday) stayed.
stays_in_week_nights	Numerical (Int)	Number of week nights (Monday to Friday) stayed.
adults	Numerical (Int)	Number of adult guests.
children	Numerical (Int)	Number of child guests.
babies	Numerical (Int)	Number of baby guests.
total_guest	Numerical (Int)	Calculated total number of guests ('adults' + 'children' + 'babies').
meal	Categorical (String)	Type of meal booked (BB, HB, FB, SC, No Meal).
country	Categorical (String)	Country of origin code (ISO 3166-1 alpha-3).
market_segment	Categorical (String)	Market segment designation (Online TA, Offline TA/TO, Direct, etc.).
distribution_channel	Categorical (String)	Booking distribution channel (TA/TO, Direct, Corporate, GDS, etc.).
is_repeated_guest	Binary (Int)	Guest loyalty flag: 1 = Repeat Guest, 0 = First-Time Guest.
previous_cancellations	Numerical (Int)	Count of previous cancellations by the customer prior to current booking.
previous_bookings_not_canceled	Numerical (Int)	Count of previous non-canceled stays by customer prior to current booking.
reserved_room_type	Categorical (String)	Code of room type reserved by guest.
assigned_room_type	Categorical (String)	Code of room type actually assigned at check-in.
booking_changes	Numerical (Int)	Number of changes/amendments made to the booking prior to check-in.
deposit_type	Categorical (String)	Deposit classification (No Deposit, Non Refundable, Refundable).
agent	Categorical / ID	ID of travel agency that made the booking.
company	Categorical / ID	ID of company/entity responsible for booking/paying.
days_in_waiting_list	Numerical (Int)	Number of days booking was on waiting list before confirmation.
customer_type	Categorical (String)	Customer booking segment (Transient, Transient-Party, Contract, Group).

Column Name	Data Type	Description
adr	Numerical (Float)	Average Daily Rate (Total lodging transactions / Total staying nights).
required_car_parking_spaces	Numerical (Int)	Number of car parking spaces requested by the guest.
total_of_special_requests	Numerical (Int)	Total count of special requests made (e.g., twin bed, high floor).
reservation_status	Categorical (String)	Final reservation state (Check-Out, Canceled, No-Show).
reservation_status_date	Datetime	Date on which the reservation status was last updated.
val_stay_length_check	Categorical (Validation)	Validation check for 0-night stays ('PASS' / 'FLAG: 0 Night Stay').
val_negative_values	Categorical (Validation)	Validation check for negative ADR values ('PASS' / 'FAIL: Negative Value').
val_cancellation_check	Categorical (Validation)	Validation check for cancellation status consistency ('PASS').
val_date_check	Categorical (Validation)	Date sanity check ('PASS' / 'FLAG: Pre-Arrival Status Date').

5. Data Cleaning & Preparation

The raw dataset was structured, audited, and cleaned in Google Sheets prior to being ingested into Google Colab for automated Python data analytics.

Google Sheets Preparation & Audit Steps

Source Preservation: The original dataset was preserved on a dedicated reference tab ([Original Dataset](#)), and all cleaning/validation workflows were performed separately on a dedicated [Cleaned Dataset](#) tab.

Data Dictionary Integration: Field definitions, expected data types, and value constraints were documented in a dedicated [Data Dictionary](#) tab to guide logical validation rules.

Data Validation & Check Columns:

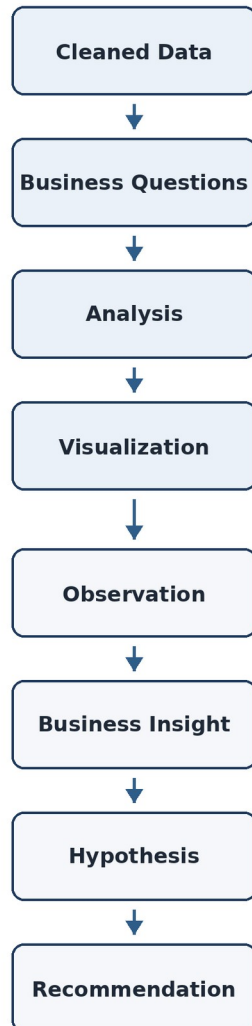
- **Stay Length Check ('val_stay_length_check'):** Audited stays where [stays_in_weekend_nights](#) + [stays_in_week_nights](#) = 0. Identified 568 records flagged as [FLAG: 0 Night Stay](#) (86,662 [PASS](#)).
- **Negative Value Check ('val_negative_values'):** Verified numerical columns for impossible negative values. Identified 1 record with a negative ADR (-6.38), while 87,229 records passed ([PASS](#)).
- **Cancellation Consistency Check ('val_cancellation_check'):** Verified logical consistency between [is_canceled](#) (1/0) and [reservation_status](#) ('Canceled', 'Check-Out', 'No-Show'). All 87,230 records passed ([PASS](#)).
- **Date Sequence Check ('val_date_check'):** Checked whether [reservation_status_date](#) occurred prior to or after the expected arrival window. Resulted in 64,985 [PASS](#) records and 22,245 [FLAG: Pre-Arrival Status Date](#) records (corresponding to early cancellations recorded on the cancellation date).
- **Country-Related Validation:** Identified 447 rows with missing, null, or placeholder country codes ([NULL](#), [Unknown](#), [TMP](#)), which were filtered out during country-specific market segment analysis without fabricating replacement values.

Ingestion into Google Colab

Following validation in Google Sheets, the cleaned dataset was loaded directly into Google Colab using `gsread` and Google OAuth authentication for processing in Python. In Colab:

- The `reservation_status_date` column was explicitly converted to a `datetime64[ns]` format.
- High-level statistical summaries (`describe()`) and categorical counts (`value_counts()`) were executed to confirm dataset integrity before evaluating business questions.

6. Analytical Approach



Prepared in Google Sheets -> analysed and visualised in Google Colab / Python

Figure 1. Analytical workflow used in the project.

Analytical Workflow

Data Ingestion: Programmatically connect Google Sheets to Python via `gsread` and import data into `pandas` DataFrames.

Business Question Framing: Address five targeted business questions sequentially using defined sub-dataframes.

Aggregations & Analytical Operations: Apply subset filtering, boolean indexing, logic categorizations, and `groupby()` aggregations to derive total counts, means, sums, and percentage metrics.

Data Visualization: Generate targeted graphical outputs using `matplotlib.pyplot` with customized layout parameters, distinct color palettes, dynamic value labels, and clear structural axis formatting.

Insights & Hypotheses Derivation: Extract objective observations directly from table outputs and visuals, construct contextual business insights, formulate non-causal behavioral hypotheses, and propose practical operational recommendations.

Tools & Python Libraries Used

- **Data Preparation:** Google Sheets
- **Environment:** Google Colab
- **Authentication:** `google.colab.auth`, `google.auth`
- **Google Sheets API Interface:** `gspread`
- **Data Manipulation & Processing:** `pandas`
- **Data Visualization:** `matplotlib.pyplot`

7. Business Question Analysis

Business Question 1: Lead Time and Cancellation Rate

Question: What is the cancellation rate for bookings with a long lead time (>90 days) compared to short lead time bookings (<30 days)?

- **Relevant Columns:** `lead_time`, `is_canceled`, `hotel`
- **Analytical Method:**
 1. Filter records to isolate short lead time bookings (`lead_time < 30`) and long lead time bookings (`lead_time > 90`), excluding intermediate lead times (30-90 days).
 2. Assign categorical labels: 'Short Lead Time (< 30 days)' and 'Long Lead Time (> 90 days)'.
 3. Aggregate metrics per group using `.groupby()`: calculate total bookings, total canceled bookings, and cancellation percentage.

Python Code Execution

Python

```
import pandas as pd
import matplotlib.pyplot as plt

# 1. Create a copy of the dataframe with only the required columns
lead_df = df[['lead_time', 'is_canceled', 'hotel']].copy()

# 2. Filter out bookings between 30 and 90 days (keeping < 30 and > 90 days)
filtered_df = lead_df[(lead_df['lead_time'] < 30) | (lead_df['lead_time'] > 90)].copy()

# 3. Assign lead-time group labels to the filtered data
filtered_df['lead_time_group'] = filtered_df['lead_time'].apply(
    lambda x: 'Short Lead Time (< 30 days)' if x < 30 else 'Long Lead Time (> 90 days)'
)

# 4. Calculate group metrics: total bookings, canceled count, and cancellation rate (%)
summary = filtered_df.groupby('lead_time_group').agg(
    total_bookings=('is_canceled', 'count'),
    canceled_bookings=('is_canceled', 'sum'),
    cancellation_rate=('is_canceled', lambda x: (x.sum() / x.count()) * 100)
).reset_index()

# Sort groups so "Short Lead Time" appears first
group_order = ['Short Lead Time (< 30 days)', 'Long Lead Time (> 90 days)']
summary['lead_time_group'] = pd.Categorical(summary['lead_time_group'], categories=group_order, ordered=True)
summary = summary.sort_values('lead_time_group').reset_index(drop=True)

# 5. Display the calculated results in a clean table format
print("=== Lead Time Cancellation Analysis Summary ===")
summary_display = summary.copy()
summary_display['cancellation_rate'] = summary_display['cancellation_rate'].map('{:.2f}%'.format)
display(summary_display)

# 6. Create the visualization comparing cancellation rates
plt.figure(figsize=(8, 6))

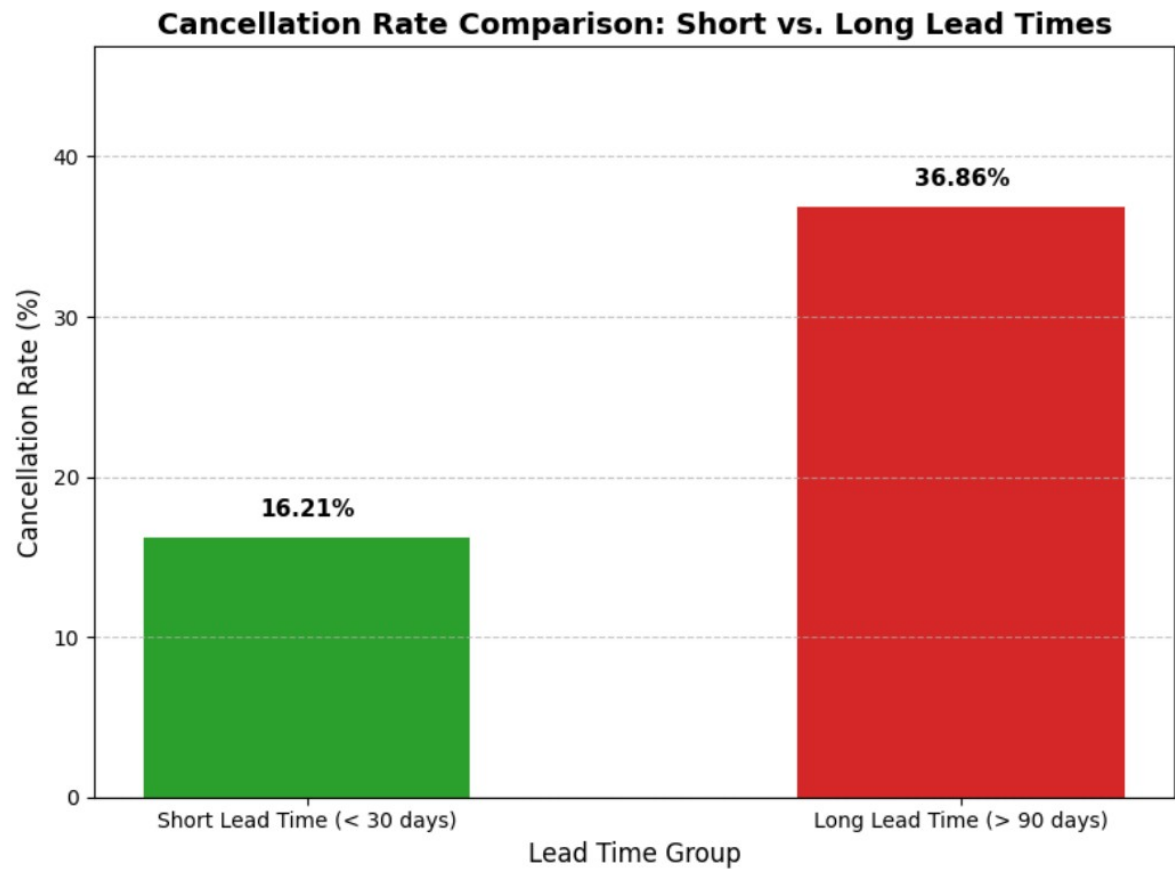
bars = plt.bar(
    summary['lead_time_group'],
    summary['cancellation_rate'],
    color=['#2ca02c', '#d62728'], # Green for short lead time, Red for long lead time
    width=0.5
)
```

```
# 7. Add percentage labels on top of each bar
for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width() / 2.0,
        height + 1,
        f'{height:.2f}%',
        ha='center',
        va='bottom',
        fontsize=11,
        fontweight='bold'
    )

# 8. Add titles, labels, and grid lines
plt.title('Cancellation Rate Comparison: Short vs. Long Lead Times', fontsize=14, fontweight='bold')
plt.xlabel('Lead Time Group', fontsize=12)
plt.ylabel('Cancellation Rate (%)', fontsize=12)
plt.ylim(0, max(summary['cancellation_rate']) + 10)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```

Analytical Output Table

lead_time_group	total_bookings	canceled_bookings	cancellation_rate
Short Lead Time (< 30 days)	33,995	5,510	16.21%
Long Lead Time (> 90 days)	29,981	11,052	36.86%



Cancellation rate comparison for short and long lead times.

- **Observation:** The cancellation rate increases from 16.21% for short lead time bookings (<30 days) to 36.86% for long lead time bookings (>90 days).
- **Business Insight:** The further out a reservation is created, the higher the likelihood that it will be canceled before arrival. Rooms held far in advance represent higher unfulfillment risk and revenue exposure for hotel inventory.

Business Question 2: Room Reassignment and ADR

Question: How frequently are reserved room types reassigned to a different room type, and how does ADR differ between reassigned and non-reassigned bookings?

- **Relevant Columns:** `reserved_room_type`, `assigned_room_type`, `adr`, `is_canceled`
- **Analytical Method:**
 1. Audit fields for missing values.
 2. Create `reassignment_status` feature by comparing `reserved_room_type` and `assigned_room_type` ('Reassigned' vs 'Not Reassigned').
 3. Calculate proportion of total bookings (including canceled bookings).
 4. Compute Average Daily Rate (ADR) exclusively on completed stays (`is_canceled == 0`) to accurately capture realized room revenue.

Python Code Execution

Python

```
import pandas as pd
import matplotlib.pyplot as plt

# 1. Select the required columns from df
room_df = df[['reserved_room_type', 'assigned_room_type', 'adr', 'is_canceled']].copy()

# 2. Handle missing/null values if present in room type or adr columns
missing_counts = room_df[['reserved_room_type', 'assigned_room_type', 'adr']].isna().sum()
print("=== Missing Value Summary ===")
print(missing_counts)
print("Note: Canceled bookings are excluded from ADR comparisons as they do not represent completed stays.")
print("\n" + "="*50 + "\n")

room_df = room_df.dropna(subset=['reserved_room_type', 'assigned_room_type', 'adr']).copy()

# 3. Create the Reassigned status label by comparing reserved vs assigned room types
room_df['reassignment_status'] = room_df.apply(
    lambda row: 'Reassigned' if row['reserved_room_type'] != row['assigned_room_type'] else 'Not Reassigned',
    axis=1
)

# 4. Calculate total bookings and percentage across ALL bookings (including canceled)
total_bookings_count = len(room_df)
booking_summary = room_df.groupby('reassignment_status').agg(
    total_bookings=('reassignment_status', 'count')
).reset_index()
booking_summary['pct_of_total_bookings'] = (booking_summary['total_bookings'] / total_bookings_count) * 100

# 5. Calculate Average ADR for completed stays only (excluding canceled bookings where is_canceled == 1)
completed_stays = room_df[room_df['is_canceled'] == 0]
adr_summary = completed_stays.groupby('reassignment_status').agg(
    avg_adr=('adr', 'mean')
).reset_index()

# 6. Merge metrics into one comprehensive summary table
summary_table = pd.merge(booking_summary, adr_summary, on='reassignment_status')

summary_table['reassignment_status'] = pd.Categorical(
    summary_table['reassignment_status'],
    categories=['Not Reassigned', 'Reassigned'],
```

```

    ordered=True
)
summary_table = summary_table.sort_values('reassignment_status').reset_index(drop=True)

# 7. Display formatted summary table
print("=== Room Reassignment Analysis Summary ===")
summary_display = summary_table.copy()
summary_display['pct_of_total_bookings'] = summary_display['pct_of_total_bookings'].map('{:.2f}%'.format)
summary_display['avg_adr'] = summary_display['avg_adr'].map('${:.2f}'.format)
display(summary_display)

# 8. Create a figure with 2 subplots (side-by-side visualizations)
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 6))

# --- Chart 1: Percentage of Bookings (Reassigned vs Not Reassigned) ---
bars1 = ax1.bar(
    summary_table['reassignment_status'],
    summary_table['pct_of_total_bookings'],
    color=['#1f77b4', '#ff7f0e'],
    width=0.4
)

for bar in bars1:
    height = bar.get_height()
    ax1.text(
        bar.get_x() + bar.get_width() / 2.0,
        height + 1,
        f'{height:.2f}%',
        ha='center',
        va='bottom',
        fontsize=11,
        fontweight='bold'
    )

ax1.set_title('Percentage of Total Bookings by Reassignment Status', fontsize=12, fontweight='bold')
ax1.set_xlabel('Reassignment Status', fontsize=11)
ax1.set_ylabel('Percentage of Total Bookings (%)', fontsize=11)
ax1.set_ylim(0, max(summary_table['pct_of_total_bookings']) + 12)
ax1.grid(axis='y', linestyle='--', alpha=0.7)

# --- Chart 2: Average ADR Comparison (Completed Stays Only) ---
bars2 = ax2.bar(
    summary_table['reassignment_status'],
    summary_table['avg_adr'],
    color=['#2ca02c', '#d62728'],
    width=0.4
)

for bar in bars2:
    height = bar.get_height()
    ax2.text(
        bar.get_x() + bar.get_width() / 2.0,
        height + 2,
        f'${height:.2f}',
        ha='center',
        va='bottom',
        fontsize=11,
        fontweight='bold'
    )

ax2.set_title('Average Daily Rate (ADR) Comparison (Completed Stays Only)', fontsize=12, fontweight='bold')
ax2.set_xlabel('Reassignment Status', fontsize=11)

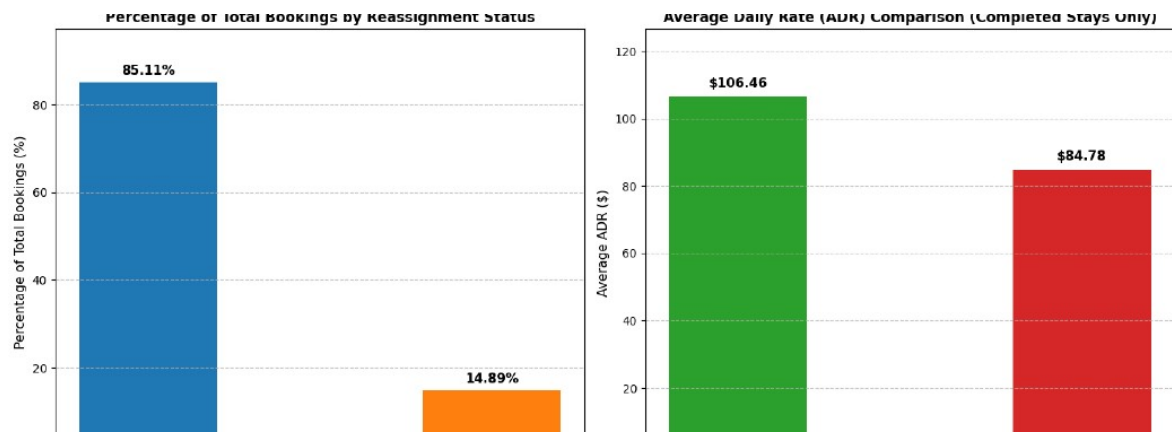
```

```
ax2.set_ylabel('Average ADR ($)', fontsize=11)
ax2.set_ylim(0, max(summary_table['avg_adr']) + 20)
ax2.grid(axis='y', linestyle='--', alpha=0.7)
```

```
plt.tight_layout()
plt.show()
```

Analytical Output Table

reassignment_status	total_bookings	pct_of_total_bookings	avg_adr
Not Reassigned	74,240	85.11%	\$106.46
Reassigned	12,990	14.89%	\$84.78



Percentage of bookings by reassignment status and average ADR for completed stays.

- **Observation:** 85.11% of all bookings (74,240) received their originally reserved room type, yielding an average ADR of \$106.46 among completed stays. Conversely, 14.89% of bookings (12,990) experienced a room type reassignment, generating a lower average ADR of \$84.78.
- **Business Insight:** Stays paying lower average daily rates coincide far more frequently with room reassignments than higher ADR stays. Guests paying standard or premium rates generally retain their reserved room categories.

Business Question 3: Special Requests, Parking and Cancellation

Question: Do bookings with special requests or parking requirements have a different cancellation rate compared with bookings with zero requests?

- **Relevant Columns:** total_of_special_requests, required_car_parking_spaces, is_canceled
- **Analytical Method:**
 1. Define two mutually exclusive categories:
 - 'No Requests or Parking': total_of_special_requests == 0 AND required_car_parking_spaces == 0.
 - 'At Least 1 Request or Parking': total_of_special_requests > 0 OR required_car_parking_spaces > 0.
 2. Aggregate group totals, canceled counts, and cancellation percentages.

Python Code Execution

Python

```
import pandas as pd
import matplotlib.pyplot as plt

# 1. Select the required columns from df
request_df = df[['total_of_special_requests', 'required_car_parking_spaces', 'is_canceled']].copy()

# 2. Check and handle missing/null values across required fields
missing_summary = request_df.isna().sum()
print("=== Missing Value Summary ===")
print(missing_summary)
print("\n" + "="*50 + "\n")

request_df = request_df.dropna().copy()

# 3. Create mutually exclusive groups
request_df['engagement_group'] = request_df.apply(
    lambda row: 'No Requests or Parking'
    if (row['total_of_special_requests'] == 0 and row['required_car_parking_spaces'] == 0)
    else 'At Least 1 Request or Parking',
    axis=1
)

# 4. Aggregate metrics for each group
summary_table = request_df.groupby('engagement_group').agg(
    total_bookings=('is_canceled', 'count'),
    canceled_bookings=('is_canceled', 'sum'),
    cancellation_rate=('is_canceled', lambda x: (x.sum() / x.count()) * 100)
).reset_index()

group_order = ['No Requests or Parking', 'At Least 1 Request or Parking']
summary_table['engagement_group'] = pd.Categorical(
    summary_table['engagement_group'],
    categories=group_order,
    ordered=True
)
summary_table = summary_table.sort_values('engagement_group').reset_index(drop=True)

# 5. Display the formatted summary table
print("=== Cancellation Rate by Requests/Parking Group ===")
summary_display = summary_table.copy()
summary_display['cancellation_rate'] = summary_display['cancellation_rate'].map('{:.2f}%'.format)
display(summary_display)
```

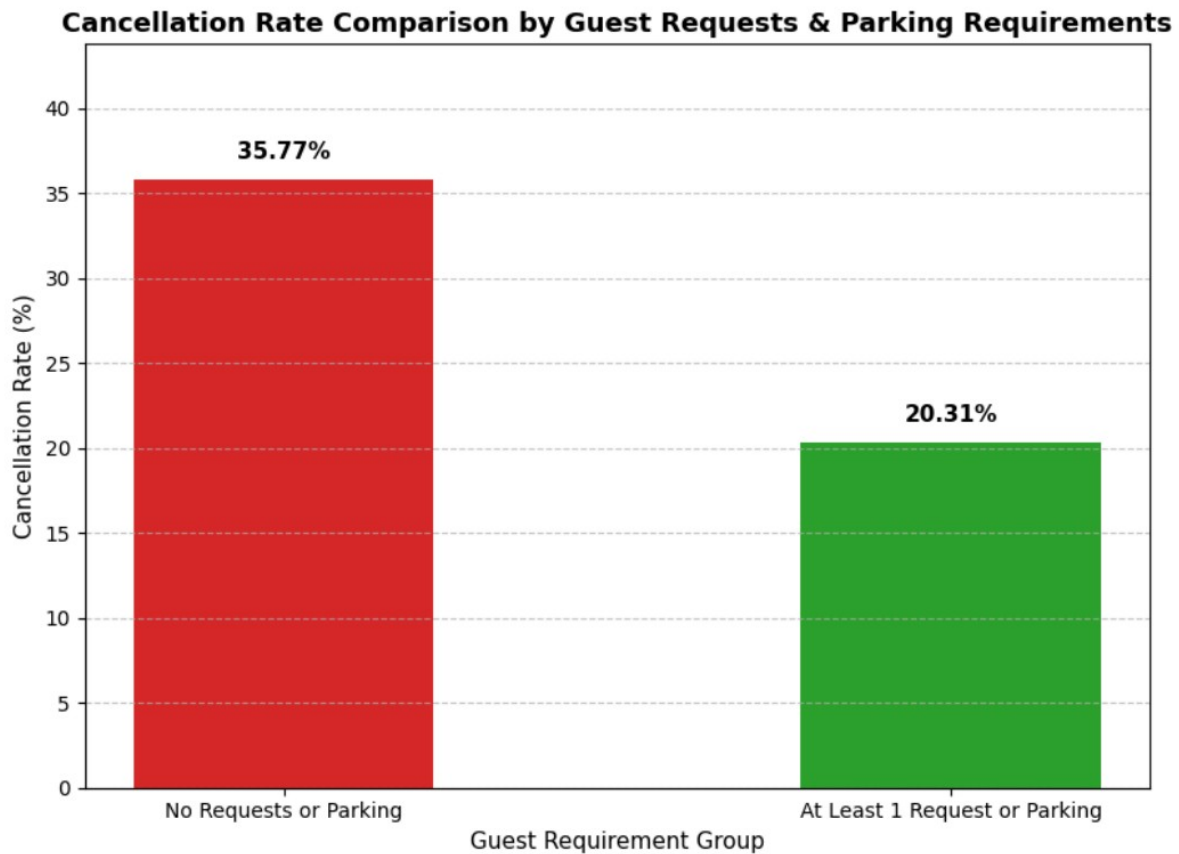
```
# 6. Create the bar chart visualization
plt.figure(figsize=(8, 6))
bars = plt.bar(
    summary_table['engagement_group'],
    summary_table['cancellation_rate'],
    color=['#d62728', '#2ca02c'],
    width=0.45
)

for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width() / 2.0,
        height + 1,
        f'{height:.2f}%',
        ha='center',
        va='bottom',
        fontsize=11,
        fontweight='bold'
    )

plt.title('Cancellation Rate Comparison by Guest Requests & Parking Requirements', fontsize=13, fontweight='bold')
plt.xlabel('Guest Requirement Group', fontsize=11)
plt.ylabel('Cancellation Rate (%)', fontsize=11)
plt.ylim(0, max(summary_table['cancellation_rate']) + 8)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```

Analytical Output Table

engagement_group	total_bookings	canceled_bookings	cancellation_rate
No Requests or Parking	46,519	14,561	35.77%
At Least 1 Request or Parking	40,711	9,448	20.31%



Cancellation rate comparison by guest requests and parking requirements.

- **Observation:** Guests with zero special requests and no parking requirements cancel at a rate of 35.77%, whereas guests submitting at least one special request or parking space cancel at 20.31%.
- **Business Insight:** Active engagement during or after booking-such as requesting specific room arrangements or parking-correlates with higher booking commitment and significantly lower cancellation risk.

Business Question 4: Repeat Guests and Booking Behaviour

Question: How do repeat guests compare with first-time guests in terms of cancellation rates and average lead times?

- **Relevant Columns:** `is_repeated_guest`, `is_canceled`, `lead_time`, `previous_bookings_not_canceled`
- **Analytical Method:**

1. Segment guests by `is_repeated_guest` ('First-Time Guest' = 0 vs 'Repeat Guest' = 1).
2. Compute total bookings, total cancellations, cancellation rate (%), mean lead time (days), and average past non-canceled stays.

Python Code Execution

Python

```
import pandas as pd
import matplotlib.pyplot as plt

# 1. Select the required columns from df
guest_df = df[['is_repeated_guest', 'is_canceled', 'lead_time', 'previous_bookings_not_canceled']].copy()

# 2. Check for missing/null values across required fields
missing_summary = guest_df.isna().sum()
print("=== Missing Value Summary ===")
print(missing_summary)
print("\n" + "="*50 + "\n")

guest_df = guest_df.dropna().copy()

# 3. Create guest category labels
guest_df['guest_category'] = guest_df['is_repeated_guest'].apply(
    lambda x: 'Repeat Guest' if x == 1 else 'First-Time Guest'
)

# 4. Aggregate metrics by guest category
summary_table = guest_df.groupby('guest_category').agg(
    total_bookings=('is_canceled', 'count'),
    canceled_bookings=('is_canceled', 'sum'),
    cancellation_rate=('is_canceled', lambda x: (x.sum() / x.count()) * 100),
    avg_lead_time=('lead_time', 'mean'),
    avg_prev_non_canceled=('previous_bookings_not_canceled', 'mean')
).reset_index()

category_order = ['First-Time Guest', 'Repeat Guest']
summary_table['guest_category'] = pd.Categorical(
    summary_table['guest_category'],
    categories=category_order,
    ordered=True
)
summary_table = summary_table.sort_values('guest_category').reset_index(drop=True)

# 5. Display the formatted summary table
print("=== Guest Loyalty Analysis Summary ===")
summary_display = summary_table.copy()
summary_display['cancellation_rate'] = summary_display['cancellation_rate'].map('{:.2f}%'.format)
summary_display['avg_lead_time'] = summary_display['avg_lead_time'].map('{:.2f} days'.format)
summary_display['avg_prev_non_canceled'] = summary_display['avg_prev_non_canceled'].map('{:.2f}'.format)
display(summary_display)

# 6. Create subplots side-by-side
```

```

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 6))

# --- Chart 1: Cancellation Rate Comparison ---
bars1 = ax1.bar(
    summary_table['guest_category'],
    summary_table['cancellation_rate'],
    color=['#1f77b4', '#2ca02c'],
    width=0.45
)

for bar in bars1:
    height = bar.get_height()
    ax1.text(
        bar.get_x() + bar.get_width() / 2.0,
        height + 1,
        f'{height:.2f}%',
        ha='center',
        va='bottom',
        fontsize=11,
        fontweight='bold'
    )

ax1.set_title('Cancellation Rate Comparison', fontsize=12, fontweight='bold')
ax1.set_xlabel('Guest Category', fontsize=11)
ax1.set_ylabel('Cancellation Rate (%)', fontsize=11)
ax1.set_ylim(0, max(summary_table['cancellation_rate']) + 8)
ax1.grid(axis='y', linestyle='--', alpha=0.7)

# --- Chart 2: Average Lead Time Comparison ---
bars2 = ax2.bar(
    summary_table['guest_category'],
    summary_table['avg_lead_time'],
    color=['#ff7f0e', '#d62728'],
    width=0.45
)

for bar in bars2:
    height = bar.get_height()
    ax2.text(
        bar.get_x() + bar.get_width() / 2.0,
        height + 2,
        f'{height:.1f} days',
        ha='center',
        va='bottom',
        fontsize=11,
        fontweight='bold'
    )

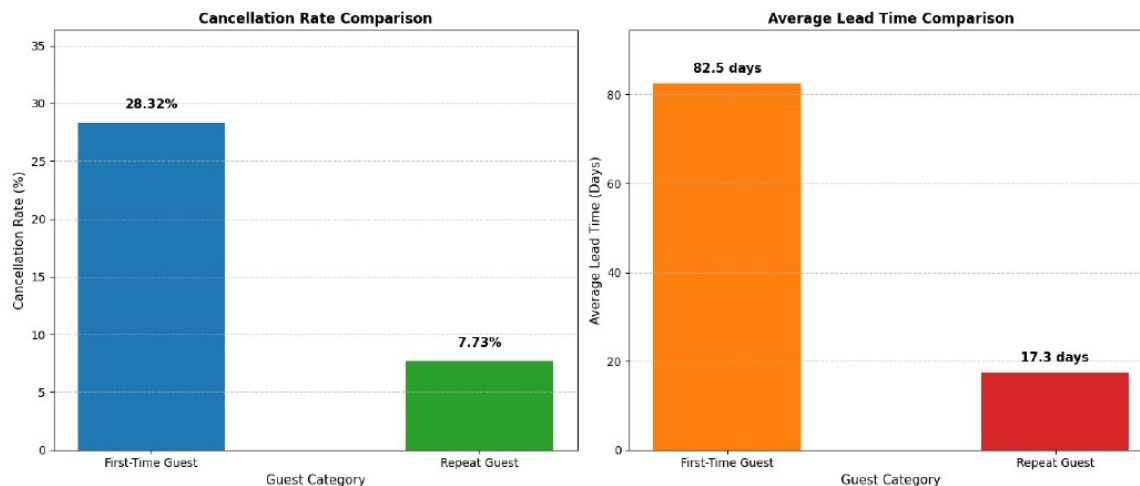
ax2.set_title('Average Lead Time Comparison', fontsize=12, fontweight='bold')
ax2.set_xlabel('Guest Category', fontsize=11)
ax2.set_ylabel('Average Lead Time (Days)', fontsize=11)
ax2.set_ylim(0, max(summary_table['avg_lead_time']) + 12)
ax2.grid(axis='y', linestyle='--', alpha=0.7)

plt.tight_layout()
plt.show()

```

Analytical Output Table

guest_category	total_bookings	canceled_bookings	cancellation_rate	avg_lead_time	avg_prev_non_canceled
First-Time Guest	83,866	23,749	28.32%	82.48 days	0.03
Repeat Guest	3,364	260	7.73%	17.33 days	4.03



Cancellation rate and average lead time comparison by guest type.

- **Observation:** First-time guests record a cancellation rate of 28.32% with an average lead time of 82.5 days. Repeat guests achieve a much lower cancellation rate of 7.73% and plan stays significantly closer to arrival (average lead time of 17.3 days).
- **Business Insight:** Returning guests present minimal cancellation risk and provide highly reliable short-term demand. First-time guests book much earlier but carry substantially higher cancellation risk.

Business Question 5: Origin Markets and Cancellation Rate

Question: Which guest origin countries have the highest cancellation rates among markets with at least 500 total bookings?

- **Relevant Columns:** `country`, `is_canceled`
- **Analytical Method:**
 1. Filter out missing, null, or placeholder country entries (447 records excluded).
 2. Aggregate total bookings, canceled count, and cancellation rate per country.
 3. Filter to include only major markets with ≥ 500 total bookings (19 eligible country markets).
 4. Sort descending by cancellation rate.

Python Code Execution

Python

```
import pandas as pd
import matplotlib.pyplot as plt

# 1. Select the required columns from df
country_df = df[['country', 'is_canceled']].copy()

# 2. Check for missing/null country values
missing_country_count = country_df['country'].isna().sum() + \
    (country_df['country'] == 'NULL').sum() + \
    (country_df['country'] == 'Unknown').sum()
print("=== Missing Country Value Summary ===")
print(f"Rows excluded due to missing/null/unknown country: {missing_country_count}")
print("\n" + "="*50 + "\n")

# Exclude missing or placeholder country values
valid_country_df = country_df[
    country_df['country'].notna() &
    (~country_df['country'].isin(['NULL', 'Unknown', 'TMP']))
].copy()

# 3. Group by country and aggregate metrics
country_summary = valid_country_df.groupby('country').agg(
    total_bookings=('is_canceled', 'count'),
    canceled_bookings=('is_canceled', 'sum'),
    cancellation_rate=('is_canceled', lambda x: (x.sum() / x.count()) * 100)
).reset_index()

# 4. Filter to include ONLY countries with at least 500 total bookings
eligible_countries = country_summary[country_summary['total_bookings'] >= 500].copy()

# 5. Sort eligible countries by cancellation rate from highest to lowest
eligible_countries = eligible_countries.sort_values(by='cancellation_rate', ascending=False).reset_index(drop=True)

# 6. Display the formatted summary table
print("=== Top Country Cancellation Rates (Min. 500 Bookings) ===")
summary_display = eligible_countries.copy()
summary_display['cancellation_rate'] = summary_display['cancellation_rate'].map('{:.2f}%'.format)
display(summary_display)

# 7. Create horizontal bar chart
chart_data = eligible_countries.sort_values(by='cancellation_rate', ascending=True).reset_index(drop=True)

plt.figure(figsize=(10, max(6, len(chart_data) * 0.4)))
```

```
bars = plt.barh(
    chart_data['country'],
    chart_data['cancellation_rate'],
    color='#d62728',
    height=0.6
)

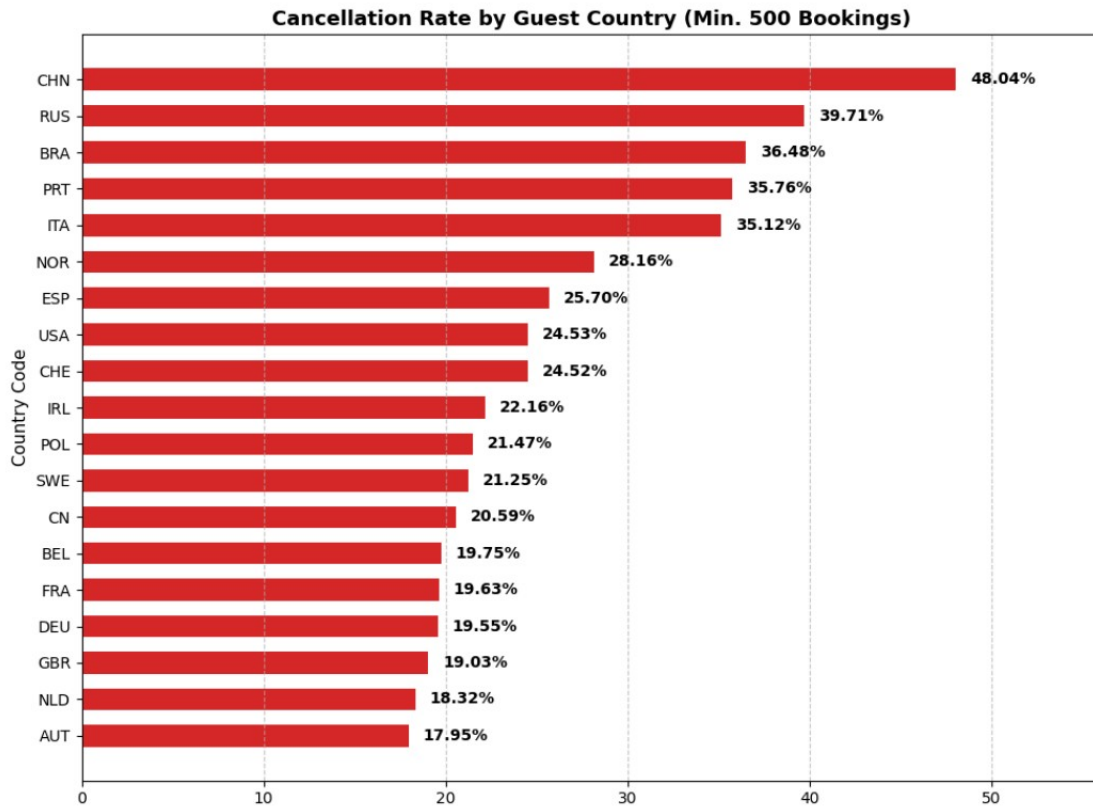
for bar in bars:
    width = bar.get_width()
    plt.text(
        width + 0.8,
        bar.get_y() + bar.get_height() / 2.0,
        f'{width:.2f}%',
        ha='left',
        va='center',
        fontsize=10,
        fontweight='bold'
    )

plt.title('Cancellation Rate by Guest Country (Min. 500 Bookings)', fontsize=13, fontweight='bold')
plt.xlabel('Cancellation Rate (%)', fontsize=11)
plt.ylabel('Country Code', fontsize=11)
plt.xlim(0, max(eligible_countries['cancellation_rate']) + 8)
plt.grid(axis='x', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```

Analytical Output Table (Min. 500 Bookings)

Rank	Country Code	Country Name / Region	total_bookings	canceled_bookings	cancellation_rate
1	CHN	China	833	400	48.04%
2	RUS	Russia	515	204	39.71%
3	BRA	Brazil	1,993	727	36.48%
4	PRT	Portugal	27,355	9,782	35.76%
5	ITA	Italy	3,061	1,075	35.12%
6	NOR	Norway	515	145	28.16%
7	ESP	Spain	7,244	1,862	25.70%
8	USA	United States	1,864	457	24.53%
9	CHE	Switzerland	1,476	362	24.52%
10	IRL	Ireland	3,015	668	22.16%
11	POL	Poland	610	131	21.47%
12	SWE	Sweden	816	173	21.25%
13	CN	China (Alt/Other)	6316	1300	20.59%
14	BEL	Belgium	2,081	411	19.75%
15	FRA	France	8,823	1,732	19.63%
16	DEU	Germany	5,385	1,053	19.55%
17	GBR	United Kingdom	10,424	1,984	19.03%
18	NLD	Netherlands	1,910	350	18.32%

Rank	Country Code	Country Name / Region	total_bookings	canceled_bookings	cancellation_rate
19	AUT	Austria	947	170	17.95%



Cancellation rate by guest country for markets with at least 500 bookings.

- **Observation:** Among origin countries with at least 500 bookings, China (CHN) exhibits the highest cancellation rate at 48.04%, followed by Russia (RUS) at 39.71% and Brazil (BRA) at 36.48%. Austria (AUT) records the lowest cancellation rate among these high-volume markets at 17.95%.
- **Business Insight:** Cancellation risk varies significantly across geographic source markets. Long-haul market segments (e.g., CHN, RUS, BRA) display higher cancellation rates than core European short-haul markets (e.g., AUT, NLD, GBR, DEU, FRA).

8. Key Insights

Lead Time Cancellation Risk: Long lead-time reservations (>90 days) experience more than double the cancellation rate (36.86%) of short lead-time reservations (<30 days; 16.21%), demonstrating that early bookings carry higher revenue vulnerability.

Room Reassignment Rate Dynamics: The majority of bookings (85.11%) stay in their reserved room type and command a higher average ADR (\$106.46) compared to reassigned bookings (14.89% of bookings, average ADR \$84.78), indicating reassignments primarily affect lower-rate stays.

Guest Engagement Predicts Completion: Guest engagement via special requests or parking space needs is associated with a substantially lower cancellation rate (20.31%) compared to non-engaged bookings (35.77%).

Repeat Guest Reliability: Repeat guests demonstrate exceptional commitment, with a 7.73% cancellation rate and an average lead time of 17.3 days, whereas first-time guests cancel at 28.32% and book 82.5 days in advance on average.

Geographic Cancellation Variance: High-volume origin markets exhibit substantial cancellation rate disparities, spanning from 17.95% (Austria) to 48.04% (China), highlighting the need for market-specific overbooking and deposit rules.

9. Hypotheses

The following hypotheses are proposed as potential explanations for the observed relationships. These represent plausible mechanisms rather than empirically proven causal claims.

Lead Time Uncertainty Hypothesis: Guests booking several months in advance face greater uncertainty regarding work schedules, personal commitments, or travel arrangements as the arrival date approaches. This increased likelihood of schedule changes results in higher cancellation rates compared to near-term bookings where travel plans are finalized.

Logistical Commitment Hypothesis: Bookings requiring parking spaces or specific accommodations often reflect trips involving family logistics, driving schedules, or specific events. Because these stays require greater prior coordination, guests have less flexibility to alter or cancel their plans.

Routine Travel Hypothesis: Returning guests frequently travel for routine purposes-such as recurring business trips or regular family visits-which are scheduled on shorter notice. Because these visits stem from established routines, they carry minimal schedule ambiguity and very low cancellation rates.

10. Recommendations

Implement Milestoned Re-Engagement Workflows for Long-Lead Bookings: Establish automated email/SMS check-ins sent at 60, 30, and 14 days prior to arrival for reservations made more than 90 days in advance. Encouraging guests to confirm itinerary details, choose stay preferences, or add amenities re-verifies intent early and provides early visibility into potential cancellations.

Target Non-Engaged Bookings with Personalization Prompts: Send pre-arrival follow-up offers to guests with zero recorded special requests or parking requirements. Offering opportunities to customize room preferences, meal add-ons, or arrival options prompts guest interaction, helps verify booking intent, and lowers cancellation likelihood.

Protect Short-Notice Inventory for Returning Guests: Allocate a dedicated portion of near-term room inventory specifically for repeat guests by offering preferred short-notice booking options or complimentary upgrades. Leveraging loyal returning guests to fill remaining inventory close to arrival secures reliable occupancy without resorting to deep public rate discounting.

11. Overall Findings / Conclusion

The project successfully evaluated 87,230 hotel booking records across five primary business areas: lead time, room reassignment, guest engagement, customer loyalty, and origin markets.

The analyses collectively demonstrate that:

- **Booking behavior varies predictably:** Cancellation risk is concentrated in long lead-time bookings, first-time guests, non-engaged bookings, and specific long-haul origin markets.

- **Engagement and loyalty drive completed stays:** Guests who interact with the hotel (special requests/parking) or have stayed previously (repeat guests) are far more likely to fulfill their reservations.
- **Pricing aligns with room stability:** Higher ADR stays experience fewer room reassignments, preserving room category assignments for higher-revenue guests.

These empirical findings provide hotel management with clear focus areas to improve demand forecasting, optimize inventory allocation, and refine pre-arrival communication strategies.

12. Limitations

Observational Scope: The analysis identifies empirical patterns and associations rather than proving direct cause-and-effect relationships.

Untested Hypotheses: The hypotheses represent plausible behavioral mechanisms but were not independently verified through controlled experiments or A/B testing.

Dataset Boundaries: The scope of analysis is constrained by the attributes and timeframes present in the provided Google Sheet/Kaggle dataset.

Actionability: The recommendations are proposed strategic interventions based on observational patterns, not pre-tested operational solutions.